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	Document Title:	RESEARCH PROPOSAL	Revision No.	00
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(1) Research Program/Project/Study Title: Modification and Optimization of Multi-Commodity Chipping Machine	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:			
Province:		Capiz and Aklan	
Region:		VI	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Agriculture/Postharvest		(8) Discipline/Field of Specialization Agricultural Engineering	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Locally and globally, multiple inventions were continuously developed to mechanize chipping. In the Philippines development were made to improve and increase production of chips of taro, saba banana, cassava, and potato and other crops for it helps provide income from local and international exports. Among the different types of chipped food products now very popular both locally and globally as a fast-food among the citizen and chip snacks industries.

Saba Banana is a triploid hybrid banana cultivar originating from the Philippines, primarily a cooking banana, it is usually boiled, steamed and fried. It is commonly used in processing banana chips and catsup. Bananas are a popular fruit that contain essential nutrients that help keep a person healthy. According to Pilipino Banana Growers and Exporters Association (PBGEA) in 2018, banana by-products such as banana chips are being exported in increasing volume to countries such as Japan, China, Korea and Middle East. However, according to Arturo. A.S. et.al on their publication on Analysis of Banana Processing Businesses & their Support Environment in the Philippines, it was found that import quality requirements of banana chips is one of the factors that hinders importation of banana chips by small-scale processing.

Root crops from taro, cassava, and potato are the main sources of 40-50% energy intakes and nutrients. These are good sources of energy, fibre, calcium, iron and vitamins. Root crops supply up to 24% of the energy required and 100% Vitamin C. Root crops are also functional foods and nutraceutical ingredients against chronic diseases helps maintain good health. Food innovations provide modern methods of cutting its either traditional or use a mechanism way. Food processing for longer continuous supply is a wise alternative. It is indeed vital to develop instant foods that are healthy and made from available natural ingredients. The department of trade and industry together with the Department of Science and Technology (DOST) has conducted a seminar training in Davao City for the exhibition for the products that have a market value like potato, taro, cassava, and sweet potato chips. This said training help the participants to expand their connection for their business not only locally but internationally.

The traditional method of chipping root crops is using knife. This method has low chipping efficiency and chipping capacity and the output are usually undesirable in terms of uniformity and thickness, notwithstanding, the tedious and cumbersome process.

An invention related to a power operated banana slicer for small scale food processing industries was developed by Sonawane et. al in 2011, (Res. Agr. Eng., 57: 144–152). It comprises of the feeding chute, push plate, push plate attachment, push rod and springs. A known disadvantage of this machine is that it is limited to either cross-sectional or longitudinal chipping only, but not both. In addition, the horizontally designed feeding chute requires a push plate with a push rod and spring to push bananas to the chipping plate thus, feeding is done on a per batch basis wherein only 6–9 bananas can be accommodated at a time. Thereby, requiring added time for loading.

Penalba, J. et al (2019) developed a hand operated chipper with feeding conveyor that produces a longitudinally sliced banana chips with a chipping capacity of 53.84 kg/hr and chipping efficiency of 92.43%. The machine has low efficiency caused by the chipped crop which sticks on the spaces between the blade and chipping plate. In addition, the machine's speed and feeding conveyor were designed for *Saba* banana chipping only.

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To invigorate the small to medium scale processing, it is important to improve their capabilities in processing activities, which is not only limited to banana but also considering other crops used as chips, as a means of obtaining varieties of products and larger share in the value chain. Accordingly, in this study, a chipping machine will be modified, optimized and evaluated for different crops use.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

This study will apply the developmental type of research, wherein an optimized chipping machine for multi-commodity use will be modified, optimized, and evaluated. The modifications will include re-designing the feeding conveyor, addition of a motor, and optimizing the speed.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The general objective of this study was to modify and optimize the existing banana chipping machine for multi-commodity use. Specifically, this study aims to:

1. modify the existing banana chipping machine for multi-commodity chipping;
2. optimize the machine in terms of the rotating speed of the chipping plate, and the speed of the feeding conveyor for commodities such as *Saba* banana, taro, cassava, and potato; and
3. perform technology valuation using R&D cost.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

A portable small-scale multi-crop chipper machine was developed in 2017 by Bello et al. which was constructed to slice different crops into choice sizes at no risk of injuries to fingers. The machine was evaluated using three selected crops; garden egg, freshly harvested cassava tuber, and plantains (ripe and unripe). The chipping efficiency, percentage crop damage and chip characteristics (length, width and thickness) were equally evaluated. Performance efficiency of the machine and chip characteristics adversely by crop moisture content. The determined efficiencies and machine capacities for chipping different crops were desirable with highest efficiency recorded in chipping unripe plantain (89.72%) while the lowest efficiency was recorded in chipping ripe plantain (48.05%). The highest machine capacity (53.58%) was recorded while chipping garden egg.

In 2018, John Prenaie E. Magpili has developed a power-driven piston type multi-crop chipper was designed to improve chipping operation of micro, small-scale and medium (MSMEs) to introduce new mechanism different from the existing conventional chippers, and develop a low cost, efficient and easy to operate and maintain chipping machine fabricated out of locally available materials. The machine performance was tested using the following parameters: feeding capacity (FC), chipping capacity (CC) and chipping efficiency (CE) as to percentage of whole and damaged chips with acceptable thickness. Outcomes of the test showed that the machine has an overall FC and CC of 139.58 kg.hr and 139.02 kg.hr, respectively. The machines have a CE of 95.48%. Whole chips obtained were 95.63% and the damaged chips were 4.37% of the total input materials. The developed machine is efficient, easy to operate and maintain. In 2011, Sonawane S.P. et al. Design and development of power operated banana slicer for small

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scale food processing industries. Res. Agr. Eng., 57: 144–152. A power operated rotary banana slicer suitable for small scale processing was designed and developed based on engineering properties of banana varieties, namely Nendran and Dwarf Cavendish. This banana slicer mainly consists of feeders for round slicing, cutter, power transmission mechanism, base support and frame. The power operated rotary slicer with three blade cutter that was operating at 360 rpm speed was developed to overcome drawbacks of existing hand or power operated rotary slicers and to meet the demand of small-scale processing industries. This slicer has slicing efficiency of about a 93–94% with effective capacity of about 100 kg/h for both varieties. The mean thickness of cut for both varieties was about 2.00 ± 0.194 mm, whereas mean roundness was of 0.84 and 0.70 for Nendran and Dwarf Cavendish varieties respectively.

In 2009, multi-crop slicing machine was designed, fabricated and evaluated for performance. The machine is powered by a three-phase, 1400 rpm, and 2 kW electric motor. The performance of the machine was evaluated in slicing four selected crops (carrot, potato, onion and yam), grouped into three sizes (small, medium and large) at five machine speeds of 39 rpm, 41 rpm, 43 rpm, 46 rpm and 48 rpm respectively. The parameters that were investigated were slicing efficiency and throughput capacity. A speed of 46 rpm was found to favour the slicing of large size crops only, while a throughput capacity of 48.9 kg/h and efficiency of 95.4% were obtained for carrot. Medium and large size samples gave good result for potato at a speed of 41 rpm with capacities and efficiencies of 72.8 kg/h, 88.9 kg/h, 97.9% and 94.8% respectively. Best result for small and medium size grade of onion we achieved at speed of 41 rpm with capacities and efficiencies of 44.6 kg/h, 71.6 kg/h, 91.7% and 96.4% respectively. A throughput capacity of 135.7 kg/h at a speed of 41 rpm and efficiency of 96% was obtained for yam. The machine is therefore observed to perform best at the optimum operating speed of 41 rpm for all the crops selected except carrot. The result of the study shows that the machine can slice root and tuber crops satisfactorily with slices ranging from 8 mm to 9 mm thickness were obtained.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Research and Sampling Design

The study is a developmental type of research. It shall include three stages such as design modification, optimization, and performance evaluation.

Design Modifications

The existing machine relates to a hand-operated banana chipping machine with feeding conveyor (Figure 1.a). It has a capacity of 53 kg/hr for longitudinal *Saba* banana chipping. The machine may be modified by:

1. adding a motor to increase the capacity and to address the issues on ease of operation;
2. adding an upper conveyor to hold and convey *Saba* banana, potato, taro, and sweet potato into the chipping chamber; and
3. adding springs to the upper and lower conveyors to allow adjustments in the loading of various materials.

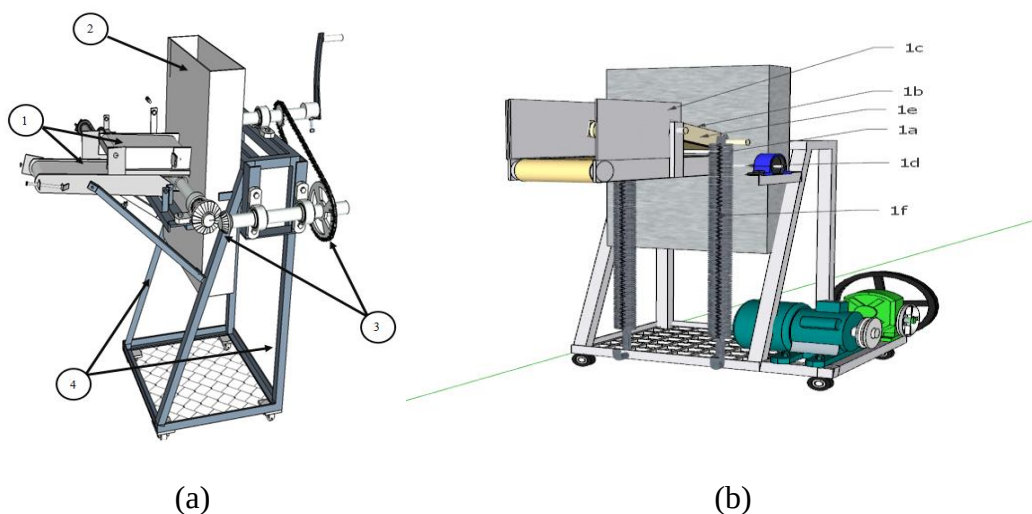


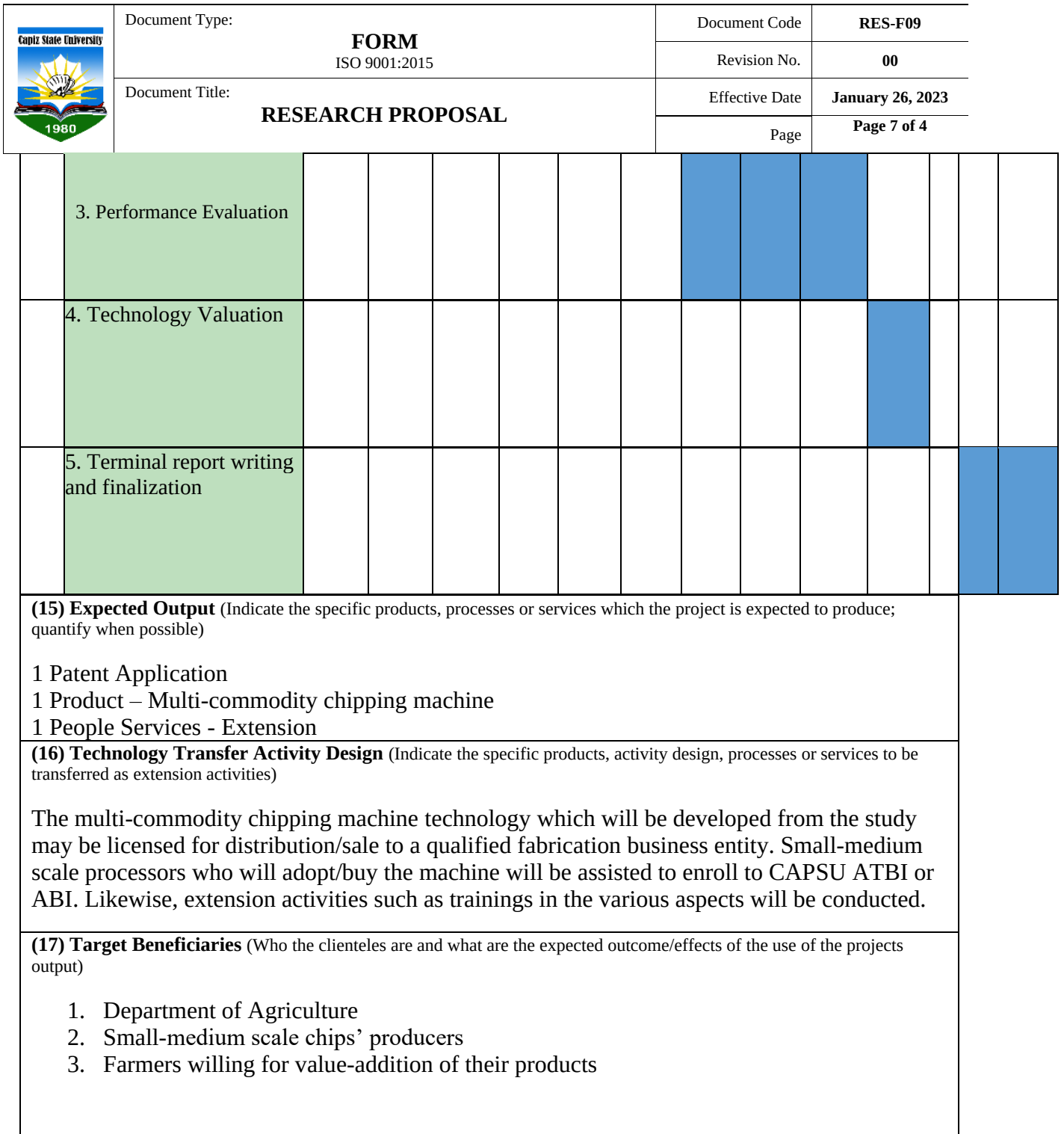
Figure 1. Perspective view of the existing banana chipping machine (a) and the proposed machine design (b)

Optimization

Machine optimization will be conducted to determine the optimum values of the chipping plate rotational speed, conveyor speed, and spring force using the Design of Experiments (DOE), specifically the Response Surface Modelling (RSM).

The experiment will use the Central Composite Design (CCD), with three factors and three levels for each factor. The arrangements in the previous machine will be used as the initial control treatment.

The experiment will be conducted in sixteen (16) runs for fifteen (15) minutes per run for each of the commodity. Data will be analyzed using the Analysis of Variance to be interpreted at 5% significance levels.



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(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Monalyn L. Oloroso	Study Leader	AB Process
Marilyn L. Bernabe	Member	Machine Development
Arlyn C. Olmo	Member	Machine Development
Ela Mae Villegas	Member	AB Process

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 3 hours per week

Program/Project Members (in hours): 3 hours per week

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
Fabricator	1 x 3 months	Php54,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Agricultural Supplies	200 kg X 5 commodities X Php20/kg	Php20,000.00
- Construction Materials	10,000.00	Php10,000.00
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		
- Printing		
C. CAPITAL OUTLAY		
Total		Phpp84,000.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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